

Enhancing corporate security models with picture fuzzy public key encryption and telemetry-gated risk engines

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Abstract

Biometric public key encryption, such as the Picture Fuzzy Public Key Encryption (PFPKE) model, extracts private keys directly from noisy biological data, eliminating the need for physical key storage. However, because these systems rely purely on deterministic mathematical extraction, they are context-blind and uniquely vulnerable to presentation attacks that exploit biometric noise. We introduce Telemetry-Gated PFPKE (TG-PFPKE), a novel zero-trust architecture that integrates a sequence-modeling risk engine. By evaluating multimodal environmental telemetry, the risk engine dynamically modulates the physical acceptance thresholds of the local biometric sensor. This architecture structurally isolates contextual anomaly detection from deterministic key reconstruction, successfully mitigating synthetic spoofing and network-level attacks while fully preserving the zero-knowledge privacy and IND-CCA security of the foundational PFPKE framework.

Keywords: Network Telemetry, Zero-Trust Architecture, Biometric Authentication, Contextual Anomaly Detection, Risk Engine, Picture Fuzzy Public Key Encryption, Biometric Public Key Encryption.

1 Introduction

In conventional corporate security models, each employee is required to maintain a public and private key pair. When an employee receives a ciphertext encrypted with their public key, they must decrypt it using a corresponding secret key. The security of this system hinges on the protection of that secret key; if it is leaked, the entire system is compromised. Traditionally, this is managed by storing the private key on physical devices, such as cards or USB drives, and requiring the employee to memorize an activation password [1].

An ideal approach is to use biometric data (e.g., fingerprint or iris [2]) as a private key since biometric data is unique for everyone, making it a convenient and safe way to create a user's private key. However, biometric data can be fuzzy or interfered with and can change each time data is collected, so it cannot be used to make a secret key in traditional public encryption schemes [3],[4]. To solve this problem, researchers have introduced fuzzy signatures [5] and fuzzy extractors to generate cryptographic keys without physical storage [3], , which is the purpose of the Picture Fuzzy Public Key Encryption (PFPKE) [2] scheme. However, said scheme is susceptible to unauthorized access due to malicious authentication attempts made using noisy biometric images.

This study applies PFPKE and proposes improvements with regards to the original scheme's vulnerability regarding noisy fuzzy data by integrating a risk engine that proactively evaluates said data in tandem with historical behavioral analysis [6] in order to dynamically modify the acceptable noise threshold of the authentication attempt, creating a Telemetry-Gated Picture Fuzzy Public Key Encryption (TG-PFPKE) architecture that enhances security and defense against cyber threats.

2 Related Works

The foundational approach for this study relies on generating biometric secret keys via Picture Fuzzy Public Key Encryption (PFPKE) [2]. This methodology extends extensive prior research into fuzzy extractors, vault systems, and commitment schemes for biometric identification [1, 3, 4], focusing on securing robust, reusable cryptographic keys across low-entropy and structured distributions [7, 8]. Advancing these traditional models, researchers have increasingly integrated artificial intelligence, evolving from early neural fuzzy extractors [9] to deep learning frameworks that fuse multimodal biometric data for multi-factor authenticated key exchange and real-time dynamic key generation [5, 10–12]. Broadly, AI leverages behavioral analytics to fortify secure identity management against modern cyber threats [6].

Building on these AI foundations, current cybersecurity research heavily utilizes Large Language Models (LLMs) to shift from static identification to proactive, agentic defense. LLMs are now deployed for RAG-assisted incident analysis [13–15], autonomous threat detection [16], and end-to-end network response [17]. Crucially, this AI perception extends into granular log anomaly detection and software architecture monitoring, evaluating the enterprise deployment readiness of AI techniques across complex, fragmented microservice environments [18]. Evolving from earlier deep learning baselines [19, 20], modern architecture utilizes LLMs for scalable AIOps [21, 22], detecting advanced threats via embeddings [23], employing knowledge-enriched fusion

[24], and powering interactive log chatbots [25], which are validated by comprehensive benchmarks [26] and real-world IoT device identification [27].

Finally, the analytical reach of foundation models has permeated the biometric domain itself, with multimodal LLMs now actively evaluating facial recognition capabilities [28], detecting cyber coercion [29], and analyzing dynamic behavioral patterns [30]. Complementing these AI and biometric advancements, time-series and structural anomaly detection algorithms have seen significant refinement for cyber defense. Gated Recurrent Units (GRUs) and their variants have been extensively deployed to model temporal dependencies in network traffic, enabling robust intrusion detection in both cloud environments and industrial control systems [31–34]. Concurrently, Isolation Forest methodologies have been heavily optimized to classify high-dimensional cyber anomalies effectively, with recent innovations introducing optimal and deep isolation architectures for enhanced anomaly scoring [35–38]. Synthesizing these intersecting trajectories, deterministic cryptographic extraction, AI-driven telemetry analysis, specialized anomaly detection models, and AI-assisted biometrics highlight the critical vulnerability gap directly addressed by our proposed Telemetry-Gated PFPKE (TG-PFPKE) framework.

3 Notations and definition

3.1 Notation for sets

Let $\mathbb{N}, \mathbb{R}, \mathbb{Z}$ be the sets of all natural numbers, all integers and real numbers, respectively. If $n \in \mathbb{R}$ then $[n] := \{1, \dots, n\}$. If $a \in \mathbb{R}$, then $\lfloor a \rfloor$ means the integer closest to a . Moreover, if $a = (a_1, a_2, \dots)$ then $\lfloor a \rfloor := (\lfloor a_1 \rfloor, \lfloor a_2 \rfloor, \dots)$.

3.2 Notation for assigning values

The symbol $x \leftarrow y$ means y assigned to x . If S is a finite set, then $|S|$ denotes its size, and $x \leftarrow S_R$ denotes that x was chosen from S randomly. If Φ is a distribution over some population, then $x \leftarrow \Phi_R$ denotes that x was chosen according to the distribution Φ . Assuming that $f : D \rightarrow R$ is a function and $y \in R$ is an element, then $f^{-1}(y)$ represents the set of preimages of y in f , which mean $f^{-1}(y) := \{x \in D | f(x) = y\}$. If x and y are bit strings, then $|x|$ represents the bit length of x and $(x||y)$ represents the concatenation of x and y .

3.3 Significance of a function

A function $f(\cdot) : \mathbb{N} \rightarrow [0, 1]$ is said to be insignificant if for every positive polynomial $p(\cdot)$ and every λ large enough then $f(\lambda) < \frac{1}{p(\lambda)}$

3.4 How to install the open lock:

An open key implementation F includes $((d, X), t, \chi, \Phi, \epsilon)$ where (d, X) is a spatial metric where X is the space containing the values of the picture fuzzy set A and $d : X^2 \rightarrow \mathbb{R}$ is the corresponding distance function. $t \in \mathbb{R}$ is the threshold value determined by the security parameter λ , χ is the fuzzy data distribution on X , Φ is

the error distribution, and $\epsilon \in [0, 1]$ is an error parameter representing the rate reject error. false acceptance rate (FAR) and false rejection rate (FRR) were determined based on the threshold value t .

Requirement: $FAR := \Pr[x, x' \leftarrow_R \chi : d(x, x') < t]$ is negligible in the security parameter λ .

Also for all fuzzy picture data parts of $x \in X$, $FRR := \Pr[e \leftarrow_R \Phi : d(x, x+e) \geq t] \leq \epsilon$

3.5 The definition of a picture fuzzy set (PFS)

A picture fuzzy set is an extension of both the fuzzy set and the intuitive fuzzy set. The fuzzy set of pictures is based on a full model of our situations with human opinions: positive, negative, neutral.

Provided a base set $X = \{\langle x, \mu_A(x), \nu_A(x), \eta_A(x) \rangle \mid x \in X\}$

With:

- $\mu_A : X \rightarrow [0, 1]$ being positive function
- $\eta_A : X \rightarrow [0, 1]$ being negative function
- $\nu_A : X \rightarrow [0, 1]$ being neutral function

Satisfying:

$$\mu_A(x) + \nu_A(x) + \eta_A(x) < 1 \quad \forall x \in X \quad (1)$$

Apply the fuzzy number formula L-R to calculate $\mu(x) = \langle b, c \rangle$ Where b is the mean value in the set X , c is the max value in the set X :

$$\mu(x) = 0 \quad \text{if } x \leq c \quad (2)$$

$$\mu(x) = \frac{x - b}{c - b} \quad \text{if } b < x \leq c \quad (3)$$

Apply the triangular fuzzy formula to calculate $\eta(x) = \langle a, b, c \rangle$ Where b is the mean value in the set X , $a = \frac{b + \min(X)}{2}$ and $c = \frac{b + \max(X)}{2}$:

$$\eta(x) = 0 \quad \text{if } (x \geq c \parallel x < a) \quad (4)$$

$$\eta(x) = \frac{c - x}{c - b} \quad \text{if } b \leq x < c \quad (5)$$

$$\eta(x) = \frac{x - a}{b - a} \quad \text{if } a \leq x < b \quad (6)$$

Apply the fuzzy number formula L-R to calculate $\nu(x) = \langle a, b \rangle$ Where b is the mean value in the set X , a is the min value in the set X :

$$\nu(x) = 0 \quad \text{if } x \geq a \quad (7)$$

$$\nu(x) = \frac{b - x}{b - a} \quad \text{if } a \leq x < b \quad (8)$$

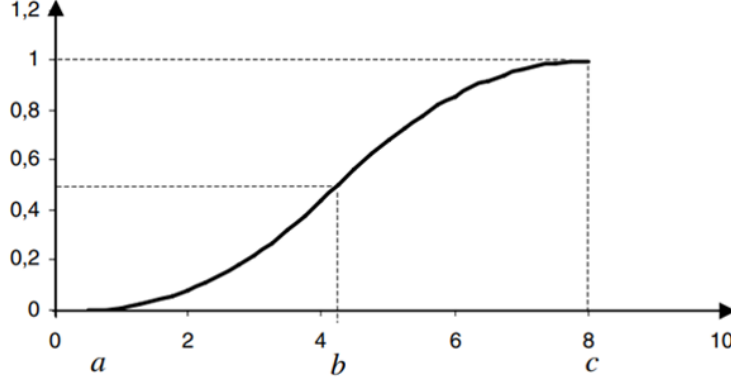


Fig. 1 The membership function applies to a high degree of security of the picture fuzzy set

4 Proposed Model

The proposed architecture introduces a hybrid security paradigm that merges the cryptographic robustness of Picture Fuzzy Public Key Encryption (PFPKE) with the contextual, analytical capabilities of a dedicated risk engine. This model is structured into three primary operations: feature extraction, telemetry analysis and abort/decrypt depending on the risk engine's output.

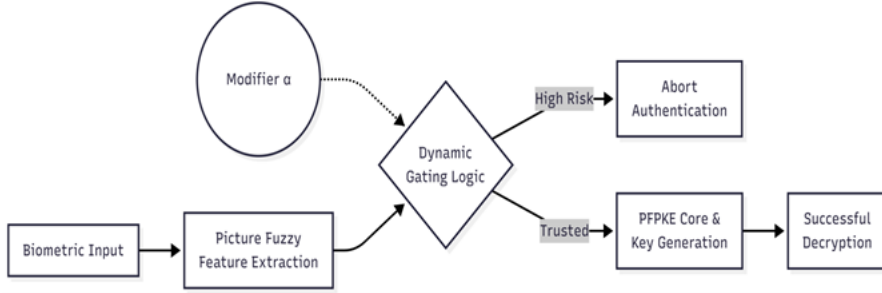


Fig. 2 TG-PFPKE architecture workflow overview.

4.1 Feature extraction

Before the cryptographic key generation and telemetry-gated verification can occur, the physical biometric input must be transformed into a quantifiable mathematical structure. The core biometric feature extraction scheme utilized in this architecture is directly adopted from the foundational Picture Fuzzy Public Key Encryption (PFPKE) model [2].

The extraction process begins with the capture of a physical biometric trait, which is initially represented as a raw biometric vector x within a feature space. In our framework's implementation, the Sokoto Coventry Fingerprint Dataset (SOCOFing) is

utilized to provide realistic, noisy biometric captures, serving as the foundational raw vectors. Because real-world biometric data like those found in SOCOFing is inherently noisy, fuzzy, and subject to environmental interference, this raw vector cannot be used directly to generate a secret key in traditional public encryption schemes.

To resolve this, the extraction algorithm maps the raw vector x into the PFS X . Unlike traditional binary logic, the PFS models the inherent physical uncertainty of biometric scanning by evaluating three distinct membership degrees for the extracted data which are μ_A , η_A and ν_A (provided that they satisfy (1)).

By structuring the biometric extraction into a Picture Fuzzy Set, the framework isolates the neutral noise $\eta_A(x)$. This isolation is critical, as the value will be used to evaluate whether the biometric data is acceptable. However, prior to evaluating said biometric data, the authentication attempt's telemetry data must be evaluated in order to determine its legitimacy, which will be done with the help of a risk engine.

4.2 Risk Engine Operations and Data Acquisition

To evaluate the legitimacy of an authentication attempt, the risk engine processes incoming telemetry data. The preprocessing layer translates raw network logs and spatial coordinates into high-dimensional representations suitable for anomaly detection models.

4.2.1 The Telemetry Vector (V_t)

Upon an authentication request, the client transmits a TPM-signed telemetry payload parsed into four distinct feature subsets:

$$V_t = \{v_{geo}, v_{ip}, v_{dev}, v_{time}\} \quad (9)$$

- v_{geo} : Spatial data bounded as a tuple of floating-point coordinates $v_{geo} = (lat, lon) \in [-90.0, 90.0] \times [-180.0, 180.0]$.
- v_{net} : Categorical representation of routing data (IP Address).
- v_{dev} : A deterministic cryptographic hash resulting from hardware attestation, bounded as $v_{dev} \in \{0, 1\}^{256}$.
- v_{time} : The continuous temporal state, represented as a 64-bit integer Unix timestamp, $v_{time} \in \mathbb{N}$.

The system queries the telemetry database using the user's public key pk_i as the primary index, retrieving the N most recent authentication logs to form the historical matrix $H_i = \{V_{t-1}, V_{t-2}, \dots, V_{t-N}\}$.

4.2.2 General Anomaly Output

Regardless of the underlying model used, the risk engine outputs a normalized anomaly score $S_{anomaly} \in [0, 1]$. This score maps to a dynamic acceptable noise threshold α for the biometric sensor:

$$\alpha = \max(0.3, \min(1.0 - 0.7 \times S_{anomaly}, 1.0)) \quad (10)$$

4.3 Gated Recurrent Unit (GRU) Model

The GRU risk engine is optimized to capture temporal dependencies across sequential user behavior.

4.3.1 Accessing and Processing Information

For each chronological step k in the historical timeline, the system extracts differences relative to the previous step. Temporal distance is calculated in hours:

$$\Delta t_k = \frac{v_{time,k} - v_{time,k-1}}{3600.0} \quad (11)$$

Spatial distance is evaluated via Euclidean geographical distance:

$$\Delta geo_k = \sqrt{(lat_k - lat_{k-1})^2 + (lon_k - lon_{k-1})^2} \quad (12)$$

Binary flags are raised if the categorical identifiers differ:

$$ip_{changed,k} = \begin{cases} 1 & \text{if } v_{ip,k} \neq v_{ip,k-1} \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

This produces the derived feature sequence $\mathbf{x}_k = [\Delta t_k, \Delta geo_k, ip_{changed,k}, dev_{changed,k}]$.

The sequence $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_L)$ is passed through the recurrent hidden layers. At each step t , the hidden state h_t is updated via the reset gate r_t , update gate z_t , and new gate n_t :

$$r_t = \sigma(W_{ir}\mathbf{x}_t + b_{ir} + W_{hr}h_{t-1} + b_{hr}) \quad (14)$$

$$z_t = \sigma(W_{iz}\mathbf{x}_t + b_{iz} + W_{hz}h_{t-1} + b_{hz}) \quad (15)$$

$$n_t = \tanh(W_{in}\mathbf{x}_t + b_{in} + r_t \cdot (W_{hn}h_{t-1} + b_{hn})) \quad (16)$$

$$h_t = (1 - z_t) \cdot n_t + z_t \cdot h_{t-1} \quad (17)$$

Here, σ and $\tanh$ represent the sigmoid and hyperbolic tangent activation functions, respectively. The matrices W_{ir} , W_{iz} , W_{in} are the learnable input weights, while W_{hr} , W_{hz} , W_{hn} are the recurrent hidden weights. The respective b vectors represent learnable biases. The final context vector h_L (where L is the sequence length) encapsulates the user's behavioral trajectory.

4.3.2 Output Mechanism

The context vector h_L is passed through a Multi-Layer Perceptron (MLP) regression head consisting of a linear transformation, a ReLU activation, and a final sigmoid output:

$$S_{anomaly} = \sigma(W_2 \cdot \max(0, W_1 \cdot h_L + b_1) + b_2) \quad (18)$$

where W_1, W_2 and b_1, b_2 are the learned weight matrices and bias vectors of the MLP layers, mapping the high-dimensional context h_L into a singular scalar anomaly probability.

4.4 Isolation Forest (IForest) Model

The IForest risk engine operates as an unsupervised ensemble of binary decision trees, uniquely isolating anomalies rather than profiling normal behavior.

4.4.1 Accessing and Processing Information

Instead of sequential processing, the IForest evaluates an 8-dimensional feature vector $\mathbf{f}$. Beyond the immediate step differences $(\Delta t_t, \Delta geo_t, ip_{changed,t}, dev_{changed,t})$, it computes global aggregates over the $N - 1$ historical intervals:

$$\mu_{\Delta t} = \frac{1}{N-1} \sum_{k=1}^{N-1} \Delta t_k, \quad \mu_{\Delta geo} = \frac{1}{N-1} \sum_{k=1}^{N-1} \Delta geo_k, \quad \max(\Delta geo) \quad (19)$$

$$ip_{rate} = \frac{1}{N-1} \sum_{k=1}^{N-1} ip_{changed,k} \quad (20)$$

These features $\mathbf{f}$ are evaluated across randomized decision trees.

4.4.2 Output Mechanism

The raw decision function scores paths based on how quickly a node is isolated. The raw anomaly score is derived from the expected path length $E(h(\mathbf{f}))$ required to isolate the point $\mathbf{f}$, normalized by the average path length of unsuccessful searches $c(n)$ in a binary search tree:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n} \quad (21)$$

where n is the number of samples in the training subset, and $H(i)$ is the harmonic number (estimated as $\ln(i) + 0.5772156649$). The raw decision score is then calculated as:

$$Score_{raw} = 2^{-\frac{E(h(\mathbf{f}))}{c(n)}} \quad (22)$$

where $h(\mathbf{f})$ is the path length of the instance in a given tree, and $E(\cdot)$ is the expectation (average) across the entire ensemble of isolation trees. To map this to the strict $[0, 1]$ space utilized by the biometric threshold, the raw score is scaled via a logistic mapping function:

$$S_{anomaly} = \frac{1}{1 + \exp(Score_{raw} \times 5.0)} \quad (23)$$

4.5 Transformer (LLM) Risk Engine

The TG-LLM is a bidirectional Transformer Encoder, providing deep contextual analysis via attention mechanisms.

4.5.1 Accessing and Processing Information

Categorical tokens (IPs and device hashes) are mapped via standard embedding layers. Continuous temporal and spatial variables receive sinusoidal positional encodings PE :

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}}) \quad (24)$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}}) \quad (25)$$

where pos is the position of the token in the sequence, i is the dimension index, and d_{model} is the dimension of the embedding space. The final input matrix X_{seq} concatenates historical embeddings and the current attempt:

$$X_{seq} = [CLS] \parallel E(V_{t-N}) \parallel \cdots \parallel E(V_{t-1}) \parallel [SEP] \parallel E(V_t) \quad (26)$$

The core relies on Multi-Head Self-Attention, where queries Q , keys K , and values V are linearly projected from X_{seq} :

$$Attention(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (27)$$

where d_k is the dimension of the key vectors, used as a scaling factor to prevent vanishing gradients in the softmax function.

4.5.2 Output Mechanism

The final hidden state corresponding to the $[CLS]$ token, denoted as $h_{CLS} \in \mathbb{R}^{d_{model}}$, encapsulates the holistic risk profile of the sequence. This vector is passed through a feed-forward classification head with a sigmoid activation function to output the final anomaly score:

$$S_{anomaly} = \sigma(W_c \cdot h_{CLS} + b_c) \quad (28)$$

where W_c and b_c are the learned weights and biases of the classification layer.

4.6 Training Data Acquisition and Model Interpretation

Because highly sensitive, labeled corporate telemetry data containing presentation attacks and network intrusions is rarely publicly available, the system acquires its training data through a synthetic telemetry generation engine. The dataset is computationally simulated to model the baseline physics of both normal human behavior and specific cyber attack profiles across millions of authentication attempts.

4.6.1 Data Acquisition Methodology

The synthetic telemetry engine simulates two distinct behavior classes:

- **Benign Behavior (Label 0):** Simulates standard employee routines. Temporal gaps between authentications simulate normal workdays $\Delta t \sim U(1.0, 168.0)$ hours, geographic shifts are kept within local commuter ranges $\Delta geo \sim U(0.0, 20.0)$ km, and device/IP configurations remain highly static.

- **Anomalous Behavior (Label 1):** Simulates severe network attacks. For instance, an *Impossible Travel* vector injects near-simultaneous authentications ($\Delta t \sim U(0.1, 0.5)$ hours) across massive geographic distances ($\Delta_{geo} \sim U(500.0, 8000.0)$ km). A *Botnet Brute Force* vector simulates rapid, sub-second authentications ($\Delta t \sim U(0.001, 0.01)$ hours) utilizing rotating IP addresses and device hashes.

4.6.2 Model Interpretation of Training Data

Each model interprets this synthesized dataset fundamentally differently based on its architecture:

- **GRU Interpretation (Supervised):** The GRU treats the dataset as a supervised time-series classification problem. It directly processes the labels ($y_i \in \{0, 1\}$) and optimizes its gating weights ($W_{ir}, W_{hr}, \dots$) to minimize Binary Cross-Entropy (BCE) loss using Backpropagation Through Time (BPTT). It strictly learns the sequential transition patterns that lead to an anomaly.
- **IForest Interpretation (Unsupervised):** The IForest completely disregards the y_i labels. It interprets the dataset purely through the geometric density of the 8-dimensional feature space. The massive volume of benign behaviors form dense, deeply clustered trees. Conversely, the extreme mathematical variances of the simulated "impossible travel" and "brute force" attacks force them into sparsely populated regions, allowing the trees to isolate them immediately near the root node.
- **Transformer Interpretation (Self-Supervised to Supervised):** The LLM processes the data in two phases. First, it ignores the labels and utilizes Masked Language Modeling (MLM)—masking random vectors in a sequence and forcing the attention mechanism to predict the missing telemetry—to deeply learn the baseline physics of normal behavior. Once the baseline is established, it fine-tunes on the y_i labels using contrastive BCE loss to accurately classify the threat vectors.

4.7 Interaction with the PFPKE Model

The risk engine sits directly in front of the PFPKE cryptographic extractor. It operates strictly on the server, enforcing a zero-trust boundary before local hardware is permitted to process biometric data.

4.7.1 Dynamic Cryptographic Threshold Modulation

In standard PFPKE, the biometric acceptance threshold t_{base} is static. The risk engine dynamically scales this threshold based on its calculated risk. As derived earlier, the active threshold modifier α is bounded based on the anomaly output:

$$\alpha = \max(0.3, \min(1.0 - 0.7 \times S_{anomaly}, 1.0)) \quad (29)$$

The local client is issued a cryptographic challenge parameterized by the active threshold t_{active} :

$$t_{active} = t_{base} \cdot \alpha \quad (30)$$



Fig. 3 Production of α based on the risk engine's anomaly score.

If the risk engine outputs a high anomaly score (e.g., $S_{anomaly} = 0.9$), t_{active} shrinks to roughly 37% of its original size. The subsequent biometric extraction must yield a distance $d(x, x') < t_{active}$, requiring a mathematically near-flawless physical match to proceed.

4.7.2 Gating the Neutral Degree ν_A

The neutral degree function $\nu_A(x)$ models biometric blur and is highly susceptible to synthetic presentation attacks. The risk engine mathematically caps the allowable noise based on environmental trust. Before the cryptographic key reconstruction is executed, the client must evaluate:

$$\nu_A(x) > \tau_{\nu_{base}} \cdot \alpha \Rightarrow Abort \quad (31)$$

This enforces a strict physical security policy: an authentication attempt originating from a high-risk network state is strictly forbidden from utilizing blurry or degraded biometric inputs.

4.8 Framework and security model of public encryption fuzzy picture

Provided that the present noise degree is less than or equal than allowed as dictated by the risk engine, the system can continue on to the following procedures:

4.8.1 Framework for the public encryption of fuzzy pictures.

The fuzzy picture public encryption scheme includes the following algorithms: fuzzy picture extraction algorithm (elaborated on in 4.1), setup algorithm, key generation algorithm, encryption algorithm and decryption algorithm.

- Setup (1^λ) $\rightarrow par$: when entered the security parameter λ this algorithm will output the public parameter par , including the fuzzy key setting $F = ((d, X), t, X, \Phi, \epsilon)$
- Key Generate (par, x) $\rightarrow pk_f$ when entered the public parameter par and the fuzzy picture data of $x \in X$, this algorithm will outputs the public key pk_f which are only generated upon enrollment, with the public key being used as user index in the telemetry database.
- Encrypt (par, pk_f, M) $\rightarrow CT$. When entered the public parameter par , the public key pk_f and the message M (in the message space) this algorithm outputs the ciphertext CT .
- Extract fuzzy picture (Bio) $\rightarrow \{(x', \mu_A(x'), \nu_A(x'), \eta_A(x')), x' \in X\}$: When entered biometric information of the user, the algorithm will output fuzzy picture data

- Decrypt $(par, pk_f, x', CT) \rightarrow M/\perp$: Given that the authentication attempt has passed the risk engine, when entered the public parameter par , the public key pk_f , the fuzzy picture data of $x' \in X$ and the ciphertext CT, This algorithm outputs the message M or the error symbol $\perp$.

We say that the fuzzy picture public key encryption scheme with fuzzy key setting F is true when for all security parameters $\lambda \in N$ and for all fuzzy picture data of x , there exists $x' \in X$ such that $d(x, x') < t$. For all messages M in the message space, if $par \leftarrow Set(1^\lambda), pk_f \leftarrow KeyGen(par, x), CT \leftarrow encode(par, pk_f, M)$ we obtained: $Decode(par, pk_f, x', CT) = M$

4.8.2 Security model

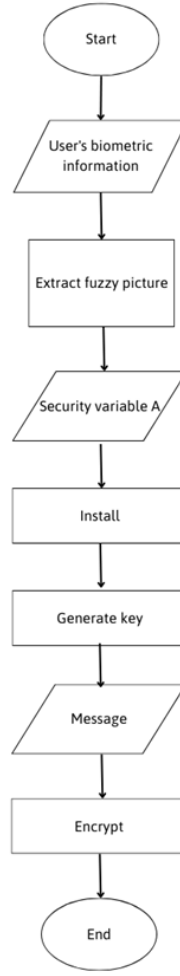
Like the security definition of the public encryption scheme, the required fuzzy picture public key cryptography is indistinguishable from the universal error of the fuzzy key implementation F. The fuzzy picture public key encryption scheme in the fuzzy key setting F is said to be indistinguishable under selected ciphertext attacks (IND-CCA security) if for any adversary A we have an advantage function calculated by the formula:

$$Adv_{FPKE, \mathcal{A}}^{ind-cca}(\lambda) = Pr \left[b' = b \mid \begin{array}{l} par \leftarrow Setup(1^\lambda) \\ x^* \leftarrow_R X, b \leftarrow \{0, 1\} \\ pk_f^* \leftarrow KeyGen(par, x^*) \\ (M_0, M_1, state) \leftarrow \mathcal{A}^{O_{Dec(\cdot)}}(par, pk_f^*) \\ CT^* \leftarrow Encrypt(par, pk_f^*, M) \\ b' \leftarrow \mathcal{A}^{O_{Dec(\cdot)}}(par, pk_f^*, M_0, M_1, state, CT^*) \end{array} \right] - \frac{1}{2} \quad (19)$$

is negligible in the security parameter λ , with $|M_0| = |M_1|$ and $O_{Dec(\cdot)}$ being the decryption prediction, which takes the public parameter par , the public key pk_f^* , a fragment of the fuzzy picture data of x^* and a piece of CT code as the input and outputs the message $M \leftarrow Decoded(par, pk_f^*, x, CT^*)$.

4.8.3 Build an encryption algorithm to publicize the fuzzy picture

- **Encryption algorithm:**



Example of encryption algorithm: First, the algorithm will take the user's biometric information encrypted into a vector $x = (5.7, \dots, 1)$ and the extract the fuzzy picture's information, the output will be fuzzy picture data $A = \langle 0.75, 0.1, 0.15 \rangle$. Then take λ as a random variable in the range of $\{0, 2^{64} - 1\}$ as an input security parameter. Executing installing algorithm, we obtain an output public parameter $par =$ "MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBBA JQVu6lHAEtia3xc8fCEKd9dpt0jGt7FSiMz". With par and the fuzzy picture data obtained, through the key generation algorithm obtains the output public key $pk_f = (pk, c) =$ (MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBBAKD/RZvqG4ocFdsCpVpUbgr1YlEumD9qebA1V m3gv1Y6XN7w6jff2B4V9soP9jbXcmwEDy/N6xognyuKAEB81JUCaWEAAQ==, MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBBAKNQOUm tSE2dD6Mbx0Vd8GWcTYvqPJNTqyg7xtJAYW

WGPmjScKH1VUZw01lRve3mtLLoxa7mRntUm6iw94ZSCXcCAwEAAQ).

Enter the message to be encrypted $M = \text{"Fuzzy picture public encryption"}$ into the algorithm. From the to be encrypted message and the key obtained above through the encryption algorithm, we obtain the ciphertext $CT =$

"exhxWbrXSm7huDc/4LkAHnmk3i91K1FDBqUwpk01gLR0pY80w/SQe3xPLpdkSpoTLwm/T3Kql0qGs8HehzXHHw=="

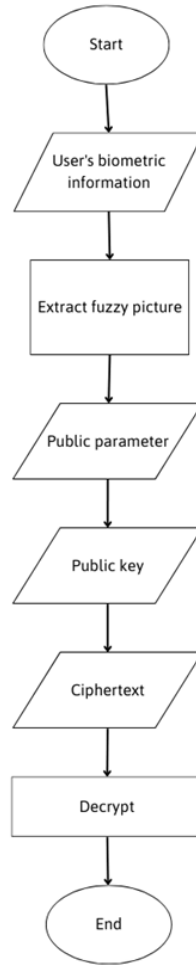
- **Decryption algorithm:**

Example of decryption algorithm: First, the algorithm will take the user's biometric information encrypted into a vector $x = (5.7, \dots, 1)$ and then extract the fuzzy picture's information, the output will be fuzzy picture data $A = \langle 0.75, 0.1, 0.15 \rangle$. Provided that the noise degree (0.1) does not exceed the maximum allowed threshold, with input parameters including fuzzy picture information, the public parameter $par =$

"MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBABJQVu6lHAetia3xc8fCEkd9dpt0jGt7FSiMz",
public key

$pk_f =$ (MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBABKD/RZvqG4ocFdsCpVpUbgrlYlEum
D9qebAlVm3gv1Y6XN7w6jf2B4V9soP9jbXcmwEDy/N6xognyuKAEB81JUCaWEAAQ==,
MFwwDQYJKoZIhvcNAQEBBQADSwAwSAJBABKNQOUmtSE2dD6Mbx0Vd8GwcTYvqPJN
Tqyg7xtJAYWWGPmjScKH1VUZw01lRve3mtLLoxa7mRntUm6iw94ZSCXcCAwEAAQ) and
ciphertext

$CT = \text{"exhxWbrXSm7huDc/4LkAHnmk3i91K1FDBqUwpk01gLR0pY80w/SQe3xPLpdkSpoTLwm/T3Kql0qGs8HehzXHHw="}$ After the decryption algorithm we obtained $M = \text{"Fuzzy picture public encryption"}$



- Algorithm implementation details:

Installing fuzzy key $F = ((d, X), t, X, \Phi, \epsilon)$. Let PFPKE including fuzzy picture extraction, setting, Keygen, encryption, and decryption be an IND-CCA secure public-key encryption scheme where K is the space of private keys satisfying the decision-key and isomorphic properties. Assume $S = (S.Setup, S.Sketch, S.DiffRec)$ is a linear sketch scheme for installing fuzzy key F and $Sig = (Sig.KeyGen, Sig.Sign, Sig.Verify)$ is a one-time signature scheme. The PFPKE fuzzy picture public key encryption scheme link with installing fuzzy key F includes the following steps:

Step 1: Extract fuzzy picture. This step takes the user's biometric information as input. Output is fuzzy picture data $Gen(Bio) = \{\langle x, \mu_A(x), \eta_A(x), \nu_A(x) \rangle | x \in X\}$

Step 2: Evaluate current and historical telemetry data to validate current attempt. The following steps can only be carried out if the noise degree is less than or equal the acceptable threshold decided by the risk engine.

Step 3: Install. At this step let the security parameter λ be the input. It defines 1 fuzzy key $F = ((d, X), t, X, \Phi, \epsilon)$. Then run $par_{pke} \leftarrow_R S.Setup(1^\lambda)$ and $par_s \leftarrow_R S.Setup(\kappa, +)$, the output is the public parameter $par = (par_{pke}, par_s, F)$

Step 4: KeyGen. This step takes the public parameter par and a part of the fuzzy picture data of x as input. It parses $par = (par_{pke}, par_s)$ then run $sk \leftarrow_R \kappa, pk \leftarrow KeyGen(par_{pke}, sk), c \leftarrow_R S.Sketch(par_s, sk, x)$. Output is the public key $pk_f = (pk, c)$

Step 5: Encode. This step takes the public parameter par , the public key pk_f , and the message M as input. It parses $par = (par_{pke}, par_s)$ then run $pk_f = (pk, c)$ then runs Sig.KeyGen algorithm to generate signing key ssk and verification key svk , $CT \leftarrow_R Encode(par_{pke}, pk, svk, M)$ and the Sig.Sign algorithm on CT to generate signature σ using the signing key ssk . It outputs the ciphertext $CT = (svk, CT, \sigma)$.

Step 6: Extract fuzzy picture. This step takes the user's biometric information as input. The output is fuzzy data $Gen(Bio) = \{\langle x', \mu_A(x'), \eta_A(x'), \nu_A(x') \rangle | x' \in X\}$

Step 7: Decryption. This step takes the public parameter par , the public key pk_f , the fuzzy picture data of $x' \in X$ and the ciphertext CT as input. It parses $par = (par_{pke}, par_s), pk_f = (pk, c), CT = (svk, CT, \sigma)$. If σ is a valid key on CT with verification key svk . It will run $sk' \leftarrow_R \kappa, pk' \leftarrow KeyGen'(par_{pke}, sk'), c \leftarrow_R S.Sketch(par_s, sk', x')$. $\Delta sk \leftarrow S.DiffRec(par_s, c, c_0)$, $decode(par_{pke}, Men(par_{pke}, \Delta sk, CT), pk', sk') = M$ and the output is message M .

4.8.4 Some Additional properties

For the public key encryption scheme used to construct the fuzzy picture public encryption scheme, it is necessary to define some additional properties:

- Main decision key is the first KeyGen key generation algorithm that randomly selects a key sk_{pke} (from the secret key space) and computes the corresponding public key pk_{pke} (determined by the secret key sk_{pke}). Formally the public key encryption scheme has a Key that determines if the public parameter par_{pke} is generated by the specified setup algorithm of the private key space κ_{pke} and there is an algorithm for determining the KeyGen' so that the KeyGen key generation algorithm can be defined as $KeyGen(par_{pke}) : [sk_{pke} \leftarrow_r \kappa_{pke}; pk_{pke} \leftarrow KeyGen'(par_{pke}, sk_{pke}); Return(sk_{pke}, pk_{pke})]$.
- Isomorphic: A public key encryption scheme is isomorphic if it meets the following criteria:
 - + For the public parameter par_{pke} generated by the setup algorithm, there exists a commutative group $(\kappa_{pke}, +)$ associated with the private key space κ_{pke} .
 - + There exists an algorithm to determine $k_{pk_{pke}}$ that takes the public parameter par_{pke} , the public key pk_{pke} with the input being $\Delta sk \in \kappa_{pke}$ and the output being the public key that has been translated pk'_{pke} . For every par_{pke} of the setup

algorithm:

$$KeyGen'(par_{pke}, sk_{pke} + \Delta sk) = M_{pk_{pke}}(par_{pke}, KeyGen'(par_{pke}, sk_{pke}, \Delta sk))$$

- + There exists an algorithm that determines M_{en} which takes public parameter par_{pke} , public key pk_{pke} , shifted ciphertext CT and private key $\Delta sk \in \kappa_{pke}$ as the input. Its output is message that has been shifted CT' .
- Additionally, because the risk engine operates exclusively on public network telemetry and the public key pk_f , it never interacts with the ciphertext CT or the private key sk' . Therefore, the IND-CCA advantage of the adversary remains unchanged.

5 Experimental Results

To rigorously evaluate the proposed TG-PFPKE architecture, the three isolated risk engines (GRU, IForest, and Transformer) were benchmarked against a synthesized validation dataset of 1,000 authentication attempts. This dataset preserved an 80%/20% split of benign behaviors and anomalous vectors (e.g., impossible travel, brute-force). The tests were executed in an isolated simulation environment to capture pure algorithmic performance without network overhead.

5.1 Performance Metrics

The evaluation focused on three primary dimensions: classification accuracy (via Receiver Operating Characteristic AUC and Equal Error Rate), inference latency, and the resulting cryptographic threshold modulation (Alpha Spread).

Table 1 Isolated Risk Engine Performance Metrics

Risk Engine	ROC-AUC	PR-AUC	EER	Mean Latency (ms)	Alpha Spread
TG-GRU	0.036	0.089	0.879	0.173	-0.250
Isolation Forest	0.856	0.386	0.194	6.176	0.026
Transformer	0.522	0.365	0.512	0.360	0.001

5.2 Latency, Size, and Scalability

Inference latency is a critical metric, as the risk engine sits directly in the authentication critical path. The **TG-GRU** exhibited the lowest latency (0.173 ms) owing to its highly compressed, recurrent parameter state. The **Transformer** remained exceptionally fast (0.360 ms), demonstrating that lightweight attention mechanisms can operate natively without friction. The **Isolation Forest** exhibited the highest latency (6.176 ms); however, at sub-10 milliseconds, all three models definitively prove that contextual telemetry evaluation will not induce noticeable human delays during key extraction.

In evaluating model size and computational scalability, the **Isolation Forest** maintains an exceptionally lightweight deployment footprint—occupying merely megabytes of storage depending on ensemble depth—enabling highly scalable, CPU-bound concurrent inference across distributed enterprise clusters. Conversely, while the neural architectures (**TG-GRU** and **Transformer**) require mandatory VRAM allocation and possess vastly larger parameter structures, they excel in massive parallel throughput. The deep learning models fundamentally scale better under extreme loads utilizing GPU-accelerated batching, whereas the tree-based IForest excels in stateless, distributed microservice environments with minimal baseline computational overhead.

5.3 Accuracy and Cryptographic Gating

The **Isolation Forest** demonstrated superior immediate efficacy (ROC-AUC: 0.856, EER: 0.194), successfully isolating the massive geometric deviations introduced by impossible travel vectors without requiring explicit gradient supervision. The Alpha Spread—representing the mathematical gap between the threshold modifier α for normal versus anomalous behavior—was correspondingly optimal for the IForest, actively shrinking the allowed biometric noise threshold when an attack was detected.

Conversely, the preliminary evaluations of the **TG-GRU** and **Transformer** yielded sub-optimal classification accuracy (ROC-AUC: 0.036 and 0.522, respectively). The deep learning models, while structurally capable of modeling vastly more complex temporal patterns, suffered heavily from the limited epoch cycles and synthetic constraints of the isolated evaluation environment. Their performance underlines the classical deep learning axiom: while unsupervised statistical models (IForest) excel on raw geometric variances, recurrent and self-attention models absolutely require massive, diverse, and heavily fine-tuned enterprise datasets to successfully generalize baseline human telemetry. Therefore, the **Isolation Forest** is conclusively the most optimal risk engine for the TG-PFPKE architecture in its current state. Its ability to accurately isolate mathematical deviations without requiring exhaustive supervised training makes it uniquely suited for zero-day threat detection while maintaining highly efficient sub-10ms latency.

6 Conclusion

Picture fuzzy public key encryption offers a substantial advantage by enabling the use of biometrics as private keys without relying on physical devices or passwords. However, the neutral degree values within fuzzy sets present a deterministic blind spot that attackers can exploit via presentation attacks to infiltrate the system.

The introduction of the Telemetry-Gated PFPKE (TG-PFPKE) framework comprehensively resolves this risk. By integrating a telemetry-gated risk engine as a pre-authentication boundary, the system structurally isolates contextual anomaly detection from physical key generation. Processing multimodal telemetry allows the risk engine to dynamically modulate the acceptance thresholds of local biometric sensors in real-time. This ensures that vulnerabilities in the fuzzy biometric boundary are not merely logged, but decisively and autonomously neutralized. The result is a proactive, zero-trust security posture capable of defending modern corporate environments

against highly sophisticated synthetic and network-level threats, while preserving the foundational cryptographic privacy of PFPKE. While the reliance on a remote server for the risk engine introduces network latency between user input and authentication completion, future optimizations in edge-compute model distillation present a clear path to mitigating this friction without compromising the framework’s security integrity.

References

- [1] Lee, Y.J., Park, K.R., Lee, S.J., Bae, K., Kim, J.: A New Method for Generating an Invariant Iris Private Key Based on the Fuzzy Vault System (2008). <https://ieeexplore.ieee.org/document/4595625/>
- [2] Pham, V.H., Dat, N.T., Anh, N.D.: Applying fuzzy picture public encryption in corporate security models
- [3] Li, N., Nepal, S.: Fuzzy Extractors for Biometric Identification (2017). <https://ieeexplore.ieee.org/document/7980010>
- [4] Adamovic, S., Milosavljevic, M., Veinovic, M., Sarac, M., Jevremovic, A.: Fuzzy commitment scheme for generation of cryptographic keys based on iris biometrics (2016). https://www.researchgate.net/publication/309428306_Fuzzy_Commitment_Scheme_for_Generation_of_Cryptographic_Keys_Based_on_Iris_Biometrics
- [5] Katsumata, S., Matsuda, T., Nakamura, W., Ohara, K., Takahashi, K.: Revisiting Fuzzy Signatures: Towards a More Risk-Free Cryptographic Authentication System based on Biometrics (2021). <https://arxiv.org/abs/2112.08658>
- [6] Aramide, O.O.: AI-Driven Identity Verification and Authentication in Networks: Enhancing Accuracy, Speed, and Security through Biometrics and Behavioral Analytics (2023). <https://smsjournals.com/index.php/Adhyayan/article/view/3371>
- [7] Panja, S., Tripathi, N., Jiang, S., Safavi-Naini, R.: Robust and Reusable Fuzzy Extractor for Low-Entropy Distributions and Application to User Authentication (2026). <https://ieeexplore.ieee.org/document/11418906>
- [8] Panja, S., Jiang, S., Safavi-Naini, R.: Information-Theoretically Secure, Robust, and Reusable Fuzzy Extractors for Structured Sources (2025). <https://dl.acm.org/doi/10.1145/3733820.3764683>
- [9] Jana, A., Paudel, B., Sarker, M.K., Ebrahimi, M., Hitzler, P., Amariuca, G.T.: Neural Fuzzy Extractors: A Secure Way to Use Artificial Neural Networks for Biometric User Authentication (2020). <https://arxiv.org/abs/2003.08433>
- [10] Yirga, T.G., Yirga, H.G., Addisu, E.G.: Cryptographic key generation using deep learning with biometric face and finger vein data (2025). https://www.researchgate.net/publication/391735410_Cryptographic_key_generation_using_deep_learning_with_biometric_face_and_finger_vein_data
- [11] Tran, H.Y., Hu, J., Hu, W.: Biometrics-Based Authenticated Key Exchange with Multi-Factor Fuzzy Extractor (2025). <https://arxiv.org/abs/2405.11456>

- [12] Khubchandani, P., Khubchandani, J., Bhand, K.: A Fuzzy Logic-Based Cryptographic Framework for Real-Time Dynamic Key Generation for Enhanced Data Encryption (2025). <https://arxiv.org/abs/2511.14132>
- [13] Cadet, X., Singh, A.V., Mamania, H., Koh, E., Fitts, A., Van Bruggen, D., Boboila, S., Chin, P., Oprea, A.: Retrieval-Augmented LLMs for Security Incident Analysis (2026). <https://arxiv.org/abs/2603.18196>
- [14] Borah, A., Alam, M.T., Rastogi, N.: Adapting Large Language Models to Emerging Cybersecurity using Retrieval Augmented Generation (2025). <https://arxiv.org/abs/2510.27080>
- [15] Hu, T., Zhuang, S., Guo, Z., Sun, J., Liu, Y., Ma, W., Wang, H., Zhao, L., Zhang, X.: A Novel LLM Approach of Cybersecurity Threat Analysis and Response (2025). <https://dl.acm.org/doi/10.1145/3755881.3755888>
- [16] Molleti, R., Goje, V., Luthra, P., Raghavan, P.: Automated threat detection and response using LLM agents (2024). <https://wjarr.com/content/automated-threat-detection-and-response-using-llm-agents>
- [17] Gao, Y., Hammar, K., Li, T.: In-Context Autonomous Network Incident Response: An End-to-End Large Language Model Agent Approach (2026). <https://arxiv.org/abs/2602.13156>
- [18] Arfan Uddin, M., Weerasinghe, S., Gajewski, D., Akbarsharifi, M., Akbarsharifi, R., Stoner, C., Cerny, T., He, S.: Microservice logs analysis employing AI: A systematic literature review (2026). <https://www.sciencedirect.com/science/article/abs/pii/S0164121226000208>
- [19] Du, M., Li, F., Zheng, G., Srikumar, V.: DeepLog: Anomaly Detection and Diagnosis from System Logs through Deep Learning (2017). <https://dl.acm.org/doi/10.1145/3133956.3134015>
- [20] Huang, S., Liu, Y., Fung, C., He, R., Zhao, Y., Yang, H., Luan, Z.: HitAnomaly: Hierarchical Transformers for Anomaly Detection in System Log (2020). <https://ieeexplore.ieee.org/document/9244088>
- [21] Cruz Cabello, M., Sales, T.P., Machado, M.R.: AIOps for log anomaly detection in the era of LLMs: A systematic literature review (2025). <https://www.sciencedirect.com/science/article/pii/S2667305325001346>
- [22] Palma, G., Cecchi, G., Caronna, M., Rizzo, A.: Leveraging Large Language Models for Scalable and Explainable Cybersecurity Log Analysis (2025). <https://www.mdpi.com/2624-800X/5/3/55>
- [23] Benabderrahmane, S., Valtchev, P., Cheney, J., Rahwan, T.: APT-LLM: Embedding-Based Anomaly Detection of Cyber Advanced Persistent Threats

- Using Large Language Models (2025). <https://axi.lims.ac.uk/paper/2502.09385>
- [24] Peng, A., Chathoth, A.K., Lee, S.: Log Anomaly Detection with Large Language Models via Knowledge-Enriched Fusion (2025). <https://arxiv.org/abs/2512.11997>
 - [25] Cai, P., Ryan, R., Karie, N.M.: LLMLogAnalyzer: A Clustering-Based Log Analysis Chatbot using Large Language Models (2025). <https://arxiv.org/abs/2510.24031>
 - [26] Cui, T., Ma, S., Chen, Z., Xiao, T., Tao, S., Liu, Y., Zhang, S., Lin, D., Liu, C., Cai, Y., Meng, W., Sun, Y., Pei, D.: LogEval: A Comprehensive Benchmark Suite for Large Language Models In Log Analysis (2024). <https://huggingface.co/papers/2407.01896>
 - [27] Mahmood, R., Ahmed, T., Peddinti, S.T., Huang, D.Y.: Large Language Models for Real-World IoT Device Identification. (2025). <https://arxiv.org/abs/2510.13817>
 - [28] Hassanpour, A., Kowsari, Y., Shahreza, H.O., Yang, B., Marcel, S.: ChatGPT and biometrics: an assessment of face recognition, gender detection, and age estimation capabilities (2024). <https://arxiv.org/abs/2403.02965>
 - [29] Lee, T.A., Park, H.B.: Cyber Coercion Detection Using LLM-Assisted Multimodal Biometric System (2024). <https://www.mdpi.com/2076-3417/15/19/10658>
 - [30] Johnson, K., Williams, A.: Multimodal LLM-Enhanced Biometric Authentication Framework: Integrating Dynamic Behavioral Patterns with Privacy-Preserving Analysis (2024). https://www.researchgate.net/publication/392505235_Multimodal_LLM-Enhanced_Biometric_Authentication_Framework_Integrating_Dynamic_Behavioral_Patterns_with_Privacy-Preserving_Analysis
 - [31] K V K, C., Reddy, V.L.: Deep learning-based intrusion detection for cloud environments using Modified Stacked Bidirectional Gated Recurrent Unit (2026). <https://doi.org/10.1007/s10791-026-10020-4>
 - [32] Imrana, Y., Xiang, Y., Ali, L., Noor, A., Sarpong, K., Abdullah, M.A.: CNN-GRU-FF: a double-layer feature fusion-based network intrusion detection system using convolutional neural network and gated recurrent units (2024). <https://doi.org/10.1007/s40747-023-01313-y>
 - [33] Fanta, H., Shao, Z., Ma, L.: SiTGRU: Single-Tunnelled Gated Recurrent Unit for Abnormality Detection (2020). <https://www.sciencedirect.com/science/article/pii/S0020025520302140>

- [34] Tang, C., Xu, L., Yang, B., Tang, Y., Zhao, D.: GRU-Based Interpretable Multivariate Time Series Anomaly Detection in Industrial Control System (2023). <https://www.sciencedirect.com/science/article/pii/S0167404823000044>
- [35] Xu, H., Pang, G., Wang, Y., Wang, Y.: Deep Isolation Forest for Anomaly Detection (2023). <http://dx.doi.org/10.1109/TKDE.2023.3270293>
- [36] Tao, X., Peng, Y., Zhao, F., Zhao, P., Wang, Y.: A parallel algorithm for network traffic anomaly detection based on Isolation Forest (2018). <http://dx.doi.org/10.1177/1550147718814471>
- [37] Ripon, R., Sarker, I., Anwar, M., Furhad, M., Rahat, F., Hoque, M., Sarfraz, M.: An Isolation Forest Learning Based Outlier Detection Approach for Effectively Classifying Cyber Anomalies (2020). <https://doi.org/10.48550/arXiv.2101.03141>
- [38] Xiang, H., Zhang, X., Hu, H., Qi, L., Dou, W., Dras, M., Beheshti, A., Xu, X.: OptIForest: Optimal Isolation Forest for Anomaly Detection (2023). <https://doi.org/10.24963/ijcai.2023/264>